

Let Time Tell: Identification and Gaussian Process Estimation for Interrupted Time Series

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September 20, 2026

Abstract

We study causal inference in interrupted time series designs where a treatment affects every unit simultaneously, so that the contemporaneous controls used by difference-in-differences and synthetic control are unavailable and the counterfactual must be extrapolated from a unit's own pre-treatment history. We establish identification within the potential outcomes framework and estimate the counterfactual by Gaussian process (GP) regression. Rather than committing to a single best-fitting trend, the GP allows for alternative trajectories compatible with the pre-treatment history and widens its intervals where extrapolation magnifies their divergence. Connecting the GP to reproducing kernel Hilbert space theory, we derive a bias decomposition that isolates the learning error associated with extrapolation and establish a worst-case bound on this error, showing how the GP posterior variance provides extrapolation-aware uncertainty quantification. The method is illustrated with calibrated simulations and an analysis of handgun purchases after the Supreme Court's *Heller* decision, a universal treatment whose practical effect concentrates in a single jurisdiction. An R package, *gpss*, implements the approach.

Keywords: causal inference, extrapolation, Gaussian process regression, interrupted time series, machine learning, reproducing kernel Hilbert space

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1 Introduction

The U.S. Supreme Court’s decision in *District of Columbia v. Heller* (2008) recognized an individual right to keep and bear firearms under the Second Amendment and overturned a 1975 D.C. law that had effectively prohibited private handgun ownership (Winkler, 2011). Did the Supreme Court ruling affect handgun purchases in D.C. and if so, by how much? Answering this question proves difficult because, although the immediate regulatory change only occurred in D.C., the ruling was a nationwide legal event that could also affect handgun purchases in other states. We therefore cannot simply regard other states as untreated controls. Thus, approaches that seek to estimate counterfactuals based on what happens in other, putatively untreated states (e.g. Difference-in-Differences (DiD) design (Callaway and Sant’Anna, 2021; Card and Krueger, 1994) and the synthetic control (SC) method (Abadie et al., 2010; Abadie and Gardeazabal, 2003)) are not applicable. Investigators may also seek to leverage an immediate before-after comparison within D.C. to identify the effect within D.C. as in the regression discontinuity in time (RDiT) design (e.g. Hausman and Rapson, 2018). However, such an approach would be an aid to identifying the effect only at the moment the event occurs, whereas we expect the decision’s impact is not entirely instantaneous, and estimating the effect over time is also of interest. We thus need a method for constructing a suitable counterfactual series absent the ruling over a wider post-ruling period, and the only available source is D.C.’s own pre-ruling history.

Many events of interest to social scientists have this structure. A war, a federal policy, or a pandemic reaches every unit at once. In this challenging but not uncommon circumstance, a natural strategy is the interrupted time series (ITS) design (Box and Tiao, 1975; Campbell and Stanley, 1963): learn the pre-event history of the target unit, project forward what would have happened, and read the effect off the gap between the projection and what actually happened. Such an extrapolation-heavy strategy faces several major threats to its validity. First, it assumes that no other event concurrent with the treatment moved the outcome. A rise in handgun purchases in D.C. cannot be attributed to the *Heller* ruling if another deregulatory policy or law took effect in the study period. Second, the projection depends on the model chosen for the pre-event history, and different models that fit the past equally well may predict very different futures. The pre-

event history alone, without a model, cannot rule out subsequent drift driven by factors unseen in that history. Third, the uncertainty of the effect estimate grows the further the projection extends beyond the event, because estimation errors are likely correlated over time and accumulate. Whatever would have moved handgun purchases in D.C. absent the ruling carries over from one month to the next, and thus a projection that is reliable for the month right after the ruling becomes far less credible years later.

A standard method for the ITS design is segmented regression, which fits a linear or polynomial trend to the whole series and adds a dummy variable for the post-event period, whose coefficient gives the effect estimate (Bernal et al., 2017). One drawback most relevant to this paper is that it commits to a single functional form for the trajectory and conditions inference on that choice. Many functions may explain the pre-event data equally well and yet diverge after the event, and the divergence amplifies with temporal distance, which we call extrapolation uncertainty. Segmented regression, and other flexible variants that commit to a single best-fitting function, fail to account for this source of uncertainty inherent in the design. Their confidence intervals may technically widen with the forecast horizon, but only conditional on the chosen functional form; they do not capture uncertainty over alternative trajectories that fit the pre-event history equally well. Adding autoregressive errors lets the interval widen somewhat with the horizon, but that widening comes from the noise around the chosen function, not from the possibility that the function itself drifts a lot in the future.

We therefore need a principled way of quantifying uncertainty that *prices* the extrapolation, so that each step beyond the data is charged by how much the plausible trajectories disagree there. This paper proposes Gaussian process (GP) regression for the ITS design, which yields such extrapolation-aware inference naturally. The reason is simple. Rather than committing to one function, the GP considers every trajectory consistent with the pre-event data. What replaces the choice of a functional form is a choice of how smooth or variable the trajectories may be, and that choice is encoded explicitly through the kernel. The projection is the average of these trajectories, and the uncertainty is their spread. Near the event they agree closely, and far from it they diverge more, so that the interval widens with the horizon by construction.

We support the proposed methodology with theoretical results and simulation studies. First,

we characterize the sources of bias that arise in the ITS design, including the gap between the best counterfactual within the assumed function class and the one the pre-event data can actually recover. Drawing on reproducing kernel Hilbert space (RKHS) theory, we derive a worst-case bound on this gap, which we call the learning error. Second, we show that this bound is extrapolation-aware and tight. When the pre-event data are noiseless, the GP posterior variance at each post-event period characterizes the largest possible disagreement between trajectories consistent with the pre-event series, given the function space induced by the kernel. In simulations calibrated to our application data, the GP intervals keep coverage near the nominal level even with short pre-event histories, whereas some competing methods lose coverage when the true trajectory drifts away from the fitted line.

Applying this method to monthly handgun background checks in D.C. and 49 other jurisdictions, we find that handgun purchases in D.C. increased immediately after the ruling by far more than any plausible continuation of the pre-ruling series allows. In contrast, we find no such effect for other jurisdictions. Furthermore, two placebo checks support this finding. Long gun purchases in D.C., which the ruling did not directly concern, show no effect, so the increase is not a general post-ruling pattern. When we move the treatment date to earlier months, the method again finds no effect, so the result is not the product of an overconfident uncertainty band either.

The rest of the paper proceeds as follows. Section 2 sets up the ITS design as a problem of extrapolation and states the identifying condition. Section 3 introduces the GP estimator and its band. Section 4 gives the bias decomposition and worst-case result behind the band. Section 5 examines finite-sample properties of the proposed estimator in calibrated simulations; Section 6 applies the method to handgun purchases after *Heller*, and Sections 7–8 situate the estimator among existing approaches and conclude.

2 Identification in Interrupted Time Series Design

To gain some intuition that informs our approach, suppose we observe a sequence of untreated outcomes before an event of interest followed by treated outcomes. Without contemporaneous controls, a researcher predicts what would have happened absent treatment by learning from pre-

treatment history. If pre-treatment outcomes follow a steady trend with a stable cycle, that pattern should continue absent treatment, and post-treatment deviations from the projection are attributed to the causal effect (Figure 1). The key question is what allows us to carry a pattern learned before treatment into the post-treatment period. Without untreated comparison units, we must rely on the relationship between outcomes and observed characteristics within each unit, and assume that this relationship remains valid when constructing the counterfactual. This section formalizes the idea.

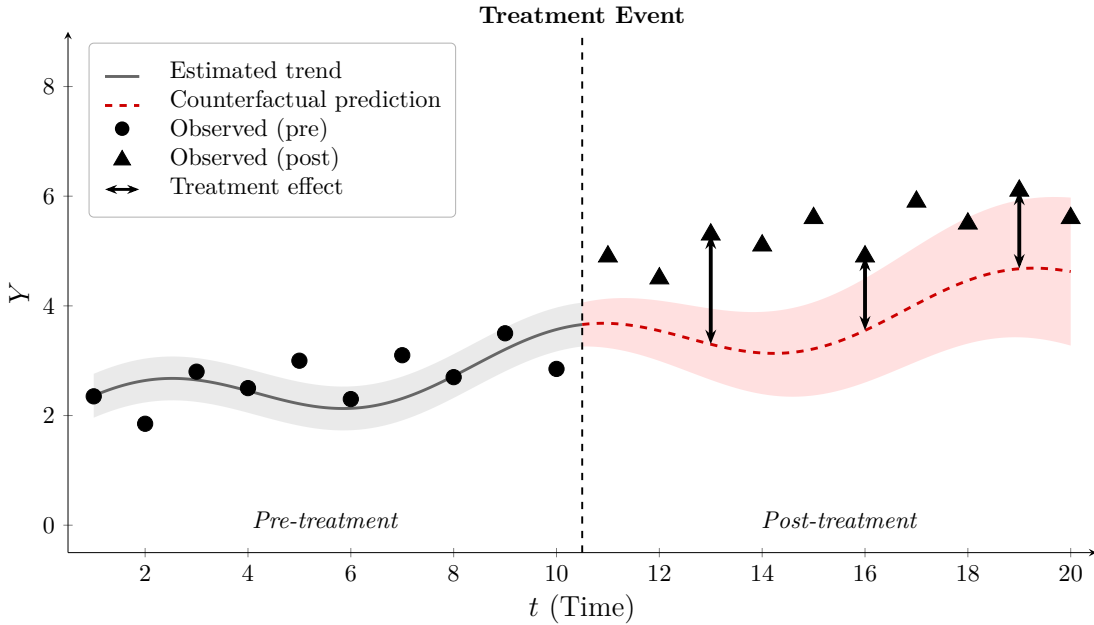


Figure 1. Illustration of Interrupted Time Series Design. *Note:* A flexible trend (solid gray) fitted to pre-treatment outcomes (black dots) is extrapolated as the counterfactual (dashed red), with 95% prediction intervals (shaded) widening with distance from the data. Treatment effects (black arrows) are the gaps between observed outcomes (black triangles) and the counterfactual; the vertical dashed line marks treatment onset.

Setup. Consider a balanced panel of N units observed over T periods. For each unit i and time t , let Y_{it} denote the scalar observed outcome, $D_{it} \in \{0, 1\}$ treatment status, and $\mathbf{X}_{it}$ a vector of observed covariates that includes time t itself alongside any additional characteristics. Unobserved time-varying attributes that may affect outcomes are collected in $\mathbf{U}_{it}$. Under the potential outcomes framework (Neyman, 1923; Rubin, 1974), let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes under treatment and control, respectively. Treatment begins simultaneously for all units at time t_0 and

persists thereafter: $D_{it} = \mathbf{1}(t \geq t_0)$.

Two standard assumptions let us interpret the pre-treatment series as untreated. We first link observed and potential outcomes through the *consistency* assumption:

$$Y_{it} = D_{it} Y_{it}(1) + (1 - D_{it}) Y_{it}(0). \quad (1)$$

To ensure that the pre-treatment observations are in fact untreated, treatment must also not act in advance of its onset (*no anticipation*):

$$Y_{it}(1) = Y_{it}(0) \quad \text{and} \quad \mathbf{X}_{it}(1) = \mathbf{X}_{it}(0), \quad t < t_0. \quad (2)$$

The first equality ensures that the outcomes observed before t_0 can be used as untreated outcomes. Together with consistency, it gives $Y_{it} = Y_{it}(0)$ for $t < t_0$. This is important because the method learns the counterfactual entirely from the pre-treatment history. If units had already changed their behavior in anticipation of treatment, that history would no longer represent the untreated potential outcomes we seek to continue. The counterfactual $Y_{it}(0)$ therefore represents what would have happened had the pre-treatment regime continued without the intervention. Expectations that were already present and remained stable may be reflected in the relationship learned from that history. What threatens no anticipation is a behavioral response to the approaching event before its formal onset. The placebo checks in Section 3.3 look for signs of pre-treatment instability, and Supplementary Material D.3 discusses this concern in each application. The second equality in (2) concerns the covariates. To evaluate the untreated counterfactual after treatment using the observed covariates, we additionally require $\mathbf{X}_{it^*}(1) = \mathbf{X}_{it^*}(0)$ for $t^* \geq t_0$. This requirement holds by construction when the covariates consist of time and calendar indicators, such as month or quarter.

Because treatment is simultaneous and universal, there are no untreated units to receive spillovers from treated units. Interactions among treated units may still affect outcomes, however. The causal quantities below are total effects inclusive of any interactions among treated units (Cerqua et al., 2023).

The distinction between the pre- and post-treatment periods determines how we use the data. Before t_0 , we observe the untreated outcomes needed to learn the counterfactual relationship. Af-

ter t_0 , we observe treated outcomes, but must predict what their untreated counterparts would have been. To make this distinction explicit, let $Y_{i,\text{pre}} = (Y_{i1}, \dots, Y_{i,t_0-1})^\top$ and $\mathbf{X}_{i,\text{pre}} = (\mathbf{X}_{i1}, \dots, \mathbf{X}_{i,t_0-1})^\top$ denote the pre-treatment outcomes and covariates. At a post-treatment period $t^* \geq t_0$, we observe Y_{it^*} and $\mathbf{X}_{it^*}$.

Structural Model. We next decompose the untreated outcome into a systematic component and idiosyncratic noise. The systematic component may depend on both observed and unobserved factors:

$$Y_{it}(0) = h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) + \varepsilon_{it}, \quad E[\varepsilon_{it} \mid \mathbf{X}_{it}, \mathbf{U}_{it}] = 0, \quad (3)$$

where the structural function $h_i : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ governs how observables and unobservables jointly determine the systematic component of outcomes, and ε_{it} is idiosyncratic noise with variance σ^2 . Note that h_i is unit-specific, allowing units exposed to the same observed and unobserved conditions to respond differently. The distinction between $\mathbf{U}_{it}$ and ε_{it} matters for counterfactual prediction. Unobserved factors that systematically affect the mean untreated outcome are represented by $\mathbf{U}_{it}$. By contrast, ε_{it} captures variation around that mean. It has mean zero conditional on both observed and unobserved factors and therefore does not itself shift the conditional mean we seek to predict. Mean-zero measurement error can also be included in ε_{it} , while systematic changes in measurement or reporting must be accounted for in the systematic component (see Supplementary Material D.3).

Target Function. Because $\mathbf{U}_{it}$ is unobserved, h_i itself is not available for prediction. The object we can hope to learn from the pre-treatment series is instead the mean untreated outcome conditional on the observables. We denote this function by $g_i : \mathcal{X} \rightarrow \mathbb{R}$ as

$$g_i(\mathbf{X}_{it}) := E[Y_{it}(0) \mid \mathbf{X}_{it}] = \int h_i(\mathbf{X}_{it}, \mathbf{u}) dP(\mathbf{U}_{it} = \mathbf{u} \mid \mathbf{X}_{it}). \quad (4)$$

Although we cannot observe $\mathbf{U}_{it}$, we can define the mean untreated outcome by averaging over its possible values. Importantly, this average is taken conditional on $\mathbf{X}_{it}$. Unobserved factors may be correlated with the observed covariates, so their distribution can differ across covariate values. Averaging over the marginal distribution instead would not generally give us the conditional mean we want. The resulting function g_i is therefore the conditional expectation that a well-specified

regression of $Y_{it}(0)$ on $\mathbf{X}_{it}$ would recover. In practice, the problem is to learn g_i from the pre-treatment history and then evaluate it at the post-treatment input $\mathbf{X}_{it^*}$. We denote the target by $g_i(\mathbf{X}_{it^*})$ and its estimate by $\hat{g}_i(\mathbf{X}_{it^*})$.

Key Identification Assumption. Defining g_i gives us a target for prediction, but it does not yet establish that the pre-treatment history is sufficient to recover the counterfactual after treatment. The main concern is that unobserved conditions, such as a concurrent event, may influence untreated outcomes in ways that the observed covariates do not capture. If those influences change, a relationship learned before treatment may no longer describe what would have happened afterward. To address this concern, we impose a restriction within each unit’s data-generating process (DGP). Once the observed covariates are known, the unobserved factors must not provide additional information about the mean untreated outcome. This is the key identifying assumption of our approach.

Assumption 1 (Mean Sufficiency). *For all units i and all periods t :*

$$\mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}, \mathbf{U}_{it}] = \mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}].$$

Mean Sufficiency does not require the unobserved factors to be constant or unrelated to the observed covariates. It requires that, once $\mathbf{X}_{it}$ is given, knowing $\mathbf{U}_{it}$ does not further change the mean untreated outcome. Individual outcomes can still fluctuate around that mean through ε_{it} . The assumption concerns the systematic part of the outcome, rather than eliminating all unexplained variation. We can see which part of the structural model this assumption restricts by adding and subtracting g_i :

$$Y_{it}(0) = \underbrace{g_i(\mathbf{X}_{it})}_{\text{observable mean}} + \underbrace{[h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) - g_i(\mathbf{X}_{it})]}_{\text{unobservable deviation}} + \underbrace{\varepsilon_{it}}_{\text{noise}}. \quad (5)$$

The middle term captures the part of the systematic outcome that remains unexplained by the observed covariates. By construction of g_i , this deviation averages to zero conditional on $\mathbf{X}_{it}$. But being zero on average does not mean that it is zero for a particular unit-period. Unobserved factors could still shift the conditional mean when their values are known. Mean Sufficiency rules out this remaining difference by requiring $h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) = g_i(\mathbf{X}_{it})$ almost surely. If the restriction fails, the

gap $h_i - g_i$ is the identification error that Section 4 isolates as Term I.

Mean Sufficiency restricts the conditional mean, not the entire distribution of untreated outcomes. Unobserved factors may still affect other features of the outcome distribution, such as its variance, as long as they do not change its conditional mean. In this sense, the assumption is weaker than the conditional-independence form of selection-on-observables (Hazlett and Xu, 2018). The analogy to selection-on-observables is useful for understanding what replaces the missing controls. In a cross-sectional setting, researchers use observed covariates to make comparisons between units. Here we rely instead on the relationship between outcomes and covariates within the same unit. We learn that relationship from the unit’s pre-treatment history and use it to construct the counterfactual after treatment. The crucial requirement is that the relationship remains valid when it is carried forward in time. Section 7 compares this identifying logic with DiD and SC in more detail.

Combining Mean Sufficiency with consistency and no anticipation gives us the regression relationship available before treatment:

$$Y_{it} = g_i(\mathbf{X}_{it}) + \varepsilon_{it}, \quad E[\varepsilon_{it} \mid \mathbf{X}_{it}] = 0, \quad t < t_0, \quad (6)$$

with the same idiosyncratic ε_{it} as in (3); under Mean Sufficiency, no additional mean-shifting component due to $\mathbf{U}_{it}$ is absorbed into it. Thus, g_i is the conditional expectation function (CEF) of the observed pre-treatment outcomes given $\mathbf{X}_{it}$. When the covariates consist of time and calendar indicators, their path is fixed by the design. Conditional on that path, the remaining variation in the untreated outcomes comes from ε_{it} , as assumed in the bias and predictive-variance calculations of Section 4.

Estimands. We define the causal estimands beginning at the unit-period level. At a post-treatment period, we observe the treated outcome but not the same unit’s realized untreated outcome. The unit- and period-specific treatment effect is $\delta_{it^*} = Y_{it^*}(1) - Y_{it^*}(0)$, the difference between the two potential outcomes for the same unit-period. Because $Y_{it^*}(0)$ is not observed, δ_{it^*} is

not identified without further assumptions. We therefore target its conditional-mean counterpart,

$$\tau_{it^*} = Y_{it^*}(1) - g_i(\mathbf{X}_{it^*}), \quad (7)$$

where $g_i(\mathbf{X}_{it^*}) = \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}]$. The estimand replaces the unrecoverable realization with its conditional mean given observables, the quantity that standard ITS methods actually estimate. Aggregate estimands average or accumulate the unit-period effects,

$$\bar{\tau}_{t^*} = \frac{1}{N} \sum_{i=1}^N \tau_{it^*}, \quad \bar{\tau}_i = \frac{1}{T - t_0 + 1} \sum_{s=0}^{T-t_0} \tau_{i,t_0+s}, \quad \tau_{i,\text{cum}} = \sum_{t^*=t_0}^T \tau_{it^*}, \quad (8)$$

the average effect at period t^* across units, the average effect for unit i over the post-treatment window, and its cumulative total. Identification of every estimand turns on recovering g_i from pre-treatment data and on its stability across t_0 , which Assumption 1 secures.

Theorem 1 (Identification of Treatment Effects). *Under consistency (1), no anticipation (2), and Assumption 1, the treatment effect at $t^* \geq t_0$ is identified as*

$$\tau_{it^*} = Y_{it^*} - g_i(\mathbf{X}_{it^*}),$$

where g_i coincides with the CEF of the observed pre-treatment data, $g_i(\mathbf{x}) = \mathbb{E}[Y_{it} \mid \mathbf{X}_{it} = \mathbf{x}]$ for $t < t_0$.

A proof is given in Supplementary Material A. Theorem 1 separates the observed and unobserved parts of the counterfactual problem. Consistency and no anticipation give us untreated outcomes before t_0 , while Mean Sufficiency restricts their systematic component to a function of the observed covariates. Consistency also allows us to replace $Y_{it^*}(1)$ with the observed post-treatment outcome Y_{it^*} . The remaining task concerns the untreated counterfactual mean $g_i(\mathbf{X}_{it^*})$. The pre-treatment distribution identifies g_i on the support of the observed inputs, but because calendar time is included in $\mathbf{X}_{it}$, evaluating the function at a post-treatment period still requires extrapolation beyond that support. This creates a further source of uncertainty: different trajectories may agree closely, or even exactly, over the observed pre-treatment periods and still diverge afterward. We refer to the range of such disagreement permitted by the assumed function class as *extrapolation uncertainty*. Section 4 formalizes this idea once the GP and its associated function space have been introduced.

The primary threat to Assumption 1 is inherent to the ITS design rather than to a particular estimator. An unobserved factor may remain stable throughout the pre-treatment period, allowing the model to fit the observed history well, but begin to influence the untreated outcome differently after treatment. If this change is not captured by the observed covariates, the resulting deviation from the projected counterfactual may be mistaken for a treatment effect. In our *Heller* application, for example, the prospect of election-related anxiety about gun restrictions motivates ending the analysis before the November election (Section 6). Such changes become harder to rule out over longer horizons. Shorter post-treatment windows are therefore often more credible, and substantive knowledge about concurrent events remains essential. The placebo checks of Section 3.3 can reveal instability in the pre-treatment period, but they cannot rule out a break that begins only after treatment.

3 Proposed Methodology: Gaussian Process Estimation for ITS

The identification result of Theorem 1 tells us what should be recovered: the counterfactual mean $g_i(\mathbf{X}_{it^*})$ at a post-treatment input; it does not yet tell us how to learn that function from the pre-treatment series and extend it beyond the observed data. This section first describes the general estimation procedure and then introduces Gaussian process (GP) regression as our proposed estimator (see [Rasmussen et al. \(2006\)](#) for a textbook treatment).

3.1 A General Estimation Procedure

Theorem 1 reduces causal identification to a function learning problem: estimate $g_i(\cdot)$ from the pre-treatment data and evaluate it at each post-treatment input. The estimation procedure follows this logic directly. We first use the pre-treatment series to learn the untreated trajectory, then extend that trajectory into the post-treatment period to form the counterfactual. The difference between the observed outcome and this counterfactual provides the estimated treatment effect.

More formally, the procedure runs in four steps. (a) For each unit i , estimate g_i from the pre-

treatment data $(\mathbf{X}_{i,\text{pre}}, Y_{i,\text{pre}})$ to obtain $\hat{g}_i$. (b) For each post-treatment period $t^* \geq t_0$, predict the counterfactual outcome $\hat{Y}_{it^*}(0) = \hat{g}_i(\mathbf{X}_{it^*})$. (c) Estimate the treatment effect $\hat{\tau}_{it^*} = Y_{it^*} - \hat{g}_i(\mathbf{X}_{it^*})$. (d) Aggregate to the summary estimands of Section 3.4 as needed. The procedure imposes no *a priori* restriction on the temporal pattern of treatment effects, such as the step, pulse, or gradual transfer functions of the standard ITS toolkit (McDowall et al., 2019; Schaffer et al., 2021); the shape of τ_{it^*} over t^* emerges from the data. Any estimator for g_i can be used for step (a) without changing the causal interpretation established in Theorem 1.

The remaining question is how well an estimator recovers g_i at a post-treatment input, where prediction requires extrapolation. We would ideally like the estimated counterfactual to equal the target on average, even when it is learned from noisy pre-treatment observations:

$$\mathbb{E}[\hat{g}_i(\mathbf{X}_{it^*}) \mid \mathbf{X}_{i,\text{pre}}, \mathbf{X}_{it^*}] = g_i(\mathbf{X}_{it^*}), \quad (9)$$

where the expectation is taken over the pre-treatment noise. We use this as a benchmark, not an assumption. With finite data and regularization, the GP may still incur learning error at post-treatment inputs. Section 4 examines how the GP posterior variance bounds this error.

3.2 Gaussian Process Estimation

Standard ITS approaches often impose a parametric trend that may not adequately capture non-linear changes in the untreated outcome. GP regression instead specifies how function values at different inputs are related through a covariance kernel. Observations close in time or similar in their covariates are assigned greater covariance, while the kernel determines how this relationship changes with distance. For our purposes, the benefit is that the GP provides both a counterfactual prediction and uncertainty about how the untreated trajectory may continue beyond the observed history. The kernel makes the assumptions behind this extrapolation explicit. We model the untreated potential outcome for unit i at time t as

$$Y_{it}(0) = f_i(\mathbf{X}_{it}) + \varepsilon_{it}, \quad f_i \sim \mathcal{GP}(0, k), \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2). \quad (10)$$

Here f_i is a unit-specific latent function, and k is the covariance kernel governing its structure. The matrix $\mathbf{K}_{\text{pre}}$, with entries $K_{tt'} = k(\mathbf{X}_{it}, \mathbf{X}_{it'})$, collects the covariances at the pre-treatment

inputs. Gaussian noise is assumed for posterior computation and hyperparameter estimation, but neither Theorem 1 nor the bias decomposition of Proposition 1 requires normality. We use the GP posterior mean $\hat{g}_i$ to estimate the target function g_i .

Posterior Prediction and Inference. We fit the GP using only the pre-treatment observations and predict the untreated outcomes at post-treatment inputs. The posterior mean gives the counterfactual estimate, while the posterior predictive covariance quantifies uncertainty about the unobserved trajectory. Under the GP model, the pre- and post-treatment outcomes are jointly Gaussian. Conditioning on the observed pre-treatment series therefore gives the following multivariate normal predictive distribution (Rasmussen et al., 2006):

$$\mathbf{Y}_{i,\text{post}}(0) \mid Y_{i,\text{pre}}, \mathbf{X}_{i,\text{pre}}, \mathbf{X}_{i,\text{post}} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \boldsymbol{\Sigma}_{\text{post}}),$$

$$\boldsymbol{\mu}_{\text{post}} = \mathbf{K}_{\text{post,pre}}(\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} Y_{i,\text{pre}}, \quad (11)$$

$$\boldsymbol{\Sigma}_{\text{post}} = \mathbf{K}_{\text{post}} + \sigma^2 I - \mathbf{K}_{\text{post,pre}}(\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{K}_{\text{pre,post}}, \quad (12)$$

where $\mathbf{K}_{\text{post,pre}}$ has (t^*, t) -entry $k(\mathbf{X}_{it^*}, \mathbf{X}_{it})$ and $\mathbf{K}_{\text{post}}$ has (t^*, t^{**}) -entry $k(\mathbf{X}_{it^*}, \mathbf{X}_{it^{**}})$. Writing $\mathbf{k}_{t^*}^\top$ for the row of $\mathbf{K}_{\text{post,pre}}$ corresponding to t^* , the corresponding entry of $\boldsymbol{\mu}_{\text{post}}$ is the GP posterior mean $\hat{g}_i(\mathbf{X}_{it^*}) = \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} Y_{i,\text{pre}}$, and the diagonal entry of $\boldsymbol{\Sigma}_{\text{post}}$ is the posterior predictive variance. Thus, the posterior mean provides the counterfactual used in the treatment-effect estimate, while the posterior predictive variance gives the corresponding uncertainty. Section 4 examines the corresponding GP variance more closely and shows how its components control the learning error.

Kernel Specification. The kernel determines how strongly the model expects the underlying function at two inputs to be correlated. In a simple time-series setting, nearby periods receive higher covariance, while observations farther apart in time receive less. The kernel therefore makes explicit what we are willing to assume about over-time comparability: different choices imply different views about how quickly the untreated trajectory may change as observations become farther apart.

As a starting point, we use the Gaussian kernel,

$$k_{\text{Gaussian}}(\mathbf{X}, \mathbf{X}') = \exp\left(-\frac{\|\mathbf{X} - \mathbf{X}'\|^2}{b}\right), \quad (13)$$

which is universal and can approximate a broad class of continuous functions (Micchelli et al., 2006). In our setting it encodes the assumption that temporally proximate observations with similar covariate values have similar function values. The same kernel also determines the function space used in the theoretical results of Section 4; we defer the RKHS interpretation until then.

Combining Kernels. Used alone, the stationary Gaussian kernel assigns little covariance to inputs far from the pre-treatment data. Its predictions therefore revert toward the prior mean, with uncertainty approaching a finite limit. This makes it less suitable for extrapolating long-run trends. For the time series considered here, we want the kernel to accommodate three common features separately: local smooth variation, recurring seasonal patterns, and a longer-run trend. Sums of valid covariance functions are valid covariance functions (Rasmussen et al., 2006), so components can be assembled to suit an application (Duvenaud et al., 2013); our working form is

$$k_{\text{combined}}(\mathbf{X}, \mathbf{X}') = k_{\text{Gaussian}}(\mathbf{X}, \mathbf{X}') + k_{\text{periodic}}(\mathbf{X}, \mathbf{X}') + k_{\text{linear}}(\mathbf{X}, \mathbf{X}'). \quad (14)$$

The periodic kernel $k_{\text{periodic}}(\mathbf{X}, \mathbf{X}') = \exp(-2 \sin^2(\pi|\mathbf{X} - \mathbf{X}'|/p) / (b/2))$ captures seasonal patterns with period p , assigning maximum covariance to time points separated by integer multiples of p (MacKay et al., 1998). The linear kernel $k_{\text{linear}}(\mathbf{X}, \mathbf{X}') = \mathbf{X}^\top \mathbf{X}'$ models trends that extend beyond the observed data, so predictive uncertainty grows with distance from the pre-treatment support. We standardize the outcome and assign unit amplitudes to the three kernel components, fixing their relative weights in the working specification (Cho et al., 2026). Where this composition is difficult to justify, researchers may report sensitivity to alternative kernel specifications.

Hyperparameter Selection. We center and scale each continuous covariate and the outcome to zero mean and unit variance within every unit, and one-hot encode categorical covariates such as the month and day-of-week indicators of our applications, so that the kernel reads all coordinates on a common scale; Supplementary Material F gives all the details. Including period indicators in $\mathbf{X}_{it}$ lets the kernel learn period-specific baselines without additional parametric mean functions to be estimated. The length-scale b controls how rapidly covariance decays with temporal distance, with larger b concentrating the prior on smooth, low-frequency functions and smaller b admitting more rapid fluctuations. We select it by the variance-maximization rule proposed in Hartman et al.

(2025), $\hat{b} = \underset{b}{\operatorname{argmax}} \mathbb{V}(K_{ij})$ for $i \neq j$, where K_{ij} are the off-diagonal entries of $\mathbf{K}_{\text{pre}}$. Intuitively, the rule avoids the two uninformative extremes: a bandwidth so small that nearly every pair of observations looks unrelated, and one so large that nearly every pair looks the same. Marginal likelihood (all hyperparameters combined) and cross-validation optimize against outcome data and risk overfitting under extrapolation, whereas this rule reads only the covariate structure, keeping the design and analysis stages separate (Chattopadhyay and Zubizarreta, 2023). Supplementary Material F re-estimates all effects across a broad range of b and reports that the findings are robust. The period p is set from the data’s natural cycle, for instance $p = 12$ for monthly data, or from the periodogram of the detrended series when unknown.

Noise Variance and Regularization. With the length-scale fixed at $\hat{b}$, the remaining parameter σ^2 controls how much of the variation in the pre-treatment series is treated as noise rather than as meaningful movement in the underlying function. Larger values allow the fitted trajectory to smooth over more of the observed variation, while smaller values make the fit follow the pre-treatment data more closely. We estimate σ^2 by maximizing the marginal likelihood, $\hat{\sigma}^2 = \underset{\sigma^2}{\operatorname{argmax}} \log p(Y_{i,\text{pre}} \mid \mathbf{X}_{i,\text{pre}}, \hat{b}, \sigma^2)$.

This same parameter also acts as a regularizer. The connection is easiest to see from the equivalent kernel ridge regression problem,

$$\hat{g}_i = \underset{f \in \mathcal{H}_k}{\operatorname{argmin}} \sum_{t < t_0} (Y_{it} - f(\mathbf{X}_{it}))^2 + \sigma^2 \|f\|_{\mathcal{H}_k}^2. \quad (15)$$

Thus, increasing σ^2 strengthens the penalty on the RKHS norm, favoring functions that fit the data less closely but are simpler under the chosen kernel. The length-scale is selected from the covariates, while σ^2 is subsequently estimated from the pre-treatment outcomes with that length-scale fixed.

3.3 Placebo Checks

Our identification strategy relies on a relationship learned before treatment to predict untreated outcomes afterward. We can examine its predictive performance using earlier pre-treatment periods as held-out targets, where the true treatment effect is zero (Eggers et al., 2023). This provides a within-unit, temporal analogue of a pre-trends check in difference-in-differences.

The procedure pretends that treatment began before its actual onset (cf. [Liu et al., 2024](#)). Take the last M pre-treatment periods $\{t_0 - M, \dots, t_0 - 1\}$ as a placebo post-treatment window, for instance matching the length of the actual one, and define relative time $m = -M, \dots, -1$, with $m = -1$ the period immediately before onset. For each m we fit the GP on all periods before $t_0 + m$ and form the placebo estimate

$$\hat{\tau}_{i,m}^{\text{placebo}} = Y_{i,t_0+m} - \hat{g}_i^{(t_0+m)}(\mathbf{X}_{i,t_0+m}), \quad m = -M, \dots, -1, \quad (16)$$

where $\hat{g}_i^{(t_0+m)}$ is the GP posterior mean trained on $\{1, \dots, t_0 + m - 1\}$. No treatment occurs at a placebo target, so the estimated effects should be near zero and their intervals should cover zero at the stated rate. Equation (16) evaluates one-step-ahead predictions, as at the first post-treatment period. Good performance here is reassuring, but it does not guarantee accuracy at longer extrapolation horizons.

The placebo is primarily a falsification check on pre-treatment forecasting performance. Systematic errors may indicate a poorly specified kernel or instability in the relationship being learned. Conversely, a clean placebo cannot establish Mean Sufficiency: an unobserved disturbance that emerges only after treatment would not be detected by this exercise. Temporal placebos can also be supplemented with placebo outcomes where available (see [Felton, 2023](#)). Supplementary Material E reports the comparisons for the empirical applications.

3.4 Aggregation for Panel Data

With N units over common periods, we fit a separate GP to each unit’s pre-treatment series and predict unit-specific counterfactuals $\hat{g}_i(\mathbf{X}_{it^*})$. Fitting unit by unit accommodates arbitrary heterogeneity in temporal trajectories, at the cost of not borrowing strength across units. The unit-specific effect estimate is $\hat{\tau}_{it^*} = Y_{it^*} - \hat{g}_i(\mathbf{X}_{it^*})$, with prediction intervals from each unit’s GP posterior predictive variance. Summary estimands aggregate the unit- and period-specific effects $\hat{\tau}_{it^*}$ over units at a fixed period, or over the post-treatment periods of a fixed unit as a mean or a cumulative total:

$$\hat{\tau}_{t^*} = \frac{1}{N} \sum_{i=1}^N \hat{\tau}_{it^*}, \quad \hat{\tau}_i = \frac{1}{T - t_0 + 1} \sum_{s=0}^{T-t_0} \hat{\tau}_{i,t_0+s}, \quad \hat{\tau}_{i,\text{cum}} = \sum_{t^*=t_0}^T \hat{\tau}_{it^*}. \quad (17)$$

Because the unit-specific GPs are fitted independently, our model treats their posterior predictive distributions as independent conditional on the data. Under this modeling choice, aggregate estimates have Gaussian posterior distributions, with uncertainty obtained by combining the relevant unit-specific predictions. Common shocks or spatial dependence may induce correlated prediction errors across units that separate GPs do not capture (cf. [Ben-Michael et al., 2023](#)). In particular, ignoring positive cross-unit dependence can produce intervals for aggregate effects that are too narrow.

4 Theoretical Properties of the Gaussian Process Estimator

The GP gives us a counterfactual prediction, but several different things can cause that prediction to deviate from the untreated outcome we would like to recover. These errors arise from distinct sources: identification, restrictions imposed by the kernel’s function class, and estimation from finite and noisy pre-treatment data. It is useful to characterize these sources before asking what the GP uncertainty band can inform us about them.

This section therefore decomposes the prediction error into four components and then focuses on the component most directly tied to extrapolation: the error in learning the counterfactual from the observed pre-treatment data. We rely on RKHS theory to show how the GP posterior variance controls this learning error.

4.1 Setup and the Four Functions

We suppress the unit subscript i throughout; all objects are unit-specific and the analysis applies unit by unit. Four functions will matter, and each corresponds to a different step between the untreated outcome and what we can actually estimate. The structural function $h(\mathbf{X}_t, \mathbf{U}_t)$ describes the systematic part of the untreated outcome when both observed and unobserved factors are known. Averaging over the unobserved factors gives $g(\mathbf{X}_t) = \mathbb{E}[Y_t(0) \mid \mathbf{X}_t]$, the counterfactual mean we would like to recover. The kernel then restricts the set of functions available to the

estimator, so we denote g^* for the best version of g available within that function class. Finally, $\hat{g}$ is what the finite pre-treatment data actually allow us to estimate.

We now define the last two objects more formally. Each kernel induces a reproducing kernel Hilbert space, $\mathcal{H}_k$, which can be understood here as the collection of functions the kernel can represent. The RKHS norm $\|f\|_{\mathcal{H}_k}$ measures how costly a function is relative to that kernel: functions that follow patterns the kernel naturally represents have smaller norms, while functions requiring less compatible behavior have larger norms. We impose a complexity budget B , restricting the model to functions whose RKHS norm does not exceed this value. A smaller budget permits fewer trajectories, while a larger one allows functions that are more costly under the kernel. The resulting function class, called an RKHS ball, is $\mathcal{F}_B = \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq B\}$. Within this function class, we define g^* as the best approximation to g over the covariate support of the full analysis period, including both pre- and post-treatment data. Let $P_{\mathbf{X}}$ denote a reference distribution over this support. Then

$$g^* = \operatorname{argmin}_{f \in \mathcal{F}_B} \|f - g\|_{L^2(P_{\mathbf{X}})}. \quad (18)$$

If g itself lies in $\mathcal{F}_B$, then $g^* = g$, and there is no approximation error. Otherwise, approximation error may remain regardless of how much pre-treatment data we observe. The technical role of the norm bound and the approximation properties of the kernel are discussed in Supplementary Material B.

The final function is $\hat{g}$, the GP posterior mean estimated from the pre-treatment data. By the representer theorem (Kimeldorf and Wahba, 1970), it lies in the empirical subspace $\mathcal{S}_{\text{pre}} = \operatorname{span}\{k(\cdot, \mathbf{X}_t) : t = 1, \dots, t_0 - 1\} \subset \mathcal{H}_k$. The pre-treatment inputs constrain the plausible function through this subspace, but they may not adequately determine its value at a post-treatment point. Finite-sample noise and regularization can introduce further learning error. Table 1 summarizes the four functions.

4.2 Bias Decomposition

The four functions above allow us to analytically ask where counterfactual prediction can go wrong. First, unobserved factors may matter in ways that are not captured by the observed covariates.

Table 1. Functions in the bias decomposition for counterfactual prediction.

Function	Definition	Governed by
$h(\mathbf{X}_t, \mathbf{U}_t)$	Structural function	Observables + unobservables
$g(\mathbf{X}_t)$	CEF: $\mathbb{E}[Y_t(0) \mid \mathbf{X}_t]$	Observables only
$g^*(\mathbf{X}_t)$	Best $\mathcal{F}_B$ approximation of g	Kernel choice
$\hat{g}(\mathbf{X}_t)$	GP estimator in $\mathcal{S}_{\text{pre}}$	Kernel + training data

Second, the function class implied by the kernel may not contain the true conditional mean. Third, even if the function class is fundamentally sound, a finite pre-treatment series may not be enough to recover the target function at a post-treatment covariate location. Finally, the realized untreated outcome contains idiosyncratic noise that no estimator can predict. These four sources correspond to identification, approximation, learning, and noise error. We begin by comparing the estimated effect with the realized treatment effect δ_{t^*} , rather than our primary estimand τ_{t^*} , to display all four sources of prediction error:

$$\hat{\tau}_{t^*} - \delta_{t^*} = Y_{t^*}(0) - \hat{g}(\mathbf{X}_{t^*}) \quad (19)$$

$$= \underbrace{[h(\mathbf{X}_{t^*}, \mathbf{U}_{t^*}) - g(\mathbf{X}_{t^*})]}_{\text{I: identification}} + \underbrace{[g(\mathbf{X}_{t^*}) - g^*(\mathbf{X}_{t^*})]}_{\text{II: approximation}} + \underbrace{[g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})]}_{\text{III: learning}} + \underbrace{\varepsilon_{t^*}}_{\text{IV: noise}}. \quad (20)$$

Our primary estimand, however, replaces the realized untreated outcome with its conditional mean. Its estimation error is $\hat{\tau}_{t^*} - \tau_{t^*} = g(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})$, which involves only Terms II and III. Terms I and IV arise when we compare the estimate with the realized effect δ_{t^*} .

Term I: Identification error. The discrepancy between $h(\mathbf{X}_{t^*}, \mathbf{U}_{t^*})$ and $g(\mathbf{X}_{t^*})$ captures the influence of unobserved factors on the systematic untreated outcome beyond what $\mathbf{X}$ captures. Under Assumption 1, $h = g$ almost surely, so Term I vanishes.

Term II: Approximation error. This is the gap between the true CEF g and the best function g^* available within the assumed function class. It is zero when $g \in \mathcal{F}_B$. If g lies outside that class, the gap remains even with unlimited pre-treatment data, because the limitation comes from the model class rather than from the sample.

Term III: Learning error. Even when $g^* = g$, the pre-treatment data may not determine g^* at a post-treatment point. The GP must learn the function from a finite pre-treatment sample, and regularization also pulls the estimate toward functions favored by the kernel (e.g., smoother

functions). This leaves a gap between the best function available in the model class and the function actually recovered from the data. Proposition 2 shows how the GP posterior variance controls this gap.

Term IV: Noise. The irreducible error ε_{t^*} has mean zero conditional on $\mathbf{X}_{t^*}, \mathbf{U}_{t^*}$ under the structural model (3) and contributes σ^2 to the outcome variance.

Under Mean Sufficiency and the stated condition on the post-treatment noise, only approximation and learning errors contribute to the conditional bias. The next proposition states this result formally.

Proposition 1 (Conditional Bias). *The conditional bias of $\hat{\tau}_{t^*}$ for the primary estimand τ_{t^*} satisfies*

$$\mathbb{E}[\hat{\tau}_{t^*} - \tau_{t^*} \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}] = \underbrace{g(\mathbf{X}_{t^*}) - g^*(\mathbf{X}_{t^*})}_{\text{approximation bias}} + \underbrace{\mathbb{E}[g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}]}_{\text{learning bias}}. \quad (21)$$

Under Assumption 1 and $\mathbb{E}[\varepsilon_{t^*} \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}] = 0$, the same expression gives the conditional bias relative to the realized effect δ_{t^*} .

The two bias terms identify two reasons why the effect estimate may remain biased even under Mean Sufficiency. Approximation bias arises when the assumed function class cannot adequately represent g ; longer pre-treatment data alone cannot resolve it. Learning bias reflects the difficulty of estimating g^* from a finite and noisy history, including the effects of regularization. It vanishes when $\hat{g}$ is unbiased for g^* , recovering the benchmark in (9) when $g^* = g$. The next proposition examines how the GP posterior variance controls learning error.

4.3 RKHS Interpretation of GP Posterior Variance

We now focus on learning error that we can control. Suppose the function class contains g , so that $g^* = g$ and approximation error is absent. The remaining question is how accurately the pre-treatment observations determine the function at a post-treatment input. Geometrically, the pre-treatment inputs span only part of the RKHS, while a post-treatment input may contain a component that this span cannot reconstruct. The size of this unrecovered component depends on how well the post-treatment input is represented by the pre-treatment inputs. The GP posterior

variance captures both this unresolved component and the uncertainty introduced by estimating the function from noisy pre-treatment outcomes.

To measure what remains undetermined by the pre-treatment inputs, we first ask how far two admissible counterfactual trajectories could diverge after treatment while agreeing at every pre-treatment input. We call this quantity extrapolation uncertainty, adapting [Shen and Meinshausen \(2025, Definition 1\)](#) to the ITS setting.

Definition 1 (Extrapolation uncertainty). *Let $\mathcal{F}_B = \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq B\}$ denote the RKHS ball with complexity budget B . For a post-treatment input $\mathbf{X}_{it^*}$, define the extrapolation uncertainty of $\mathcal{F}_B$ as*

$$\omega_B(\mathbf{X}_{t^*}) = \sup\{|f(\mathbf{X}_{t^*}) - f'(\mathbf{X}_{t^*})| : f, f' \in \mathcal{F}_B, f(\mathbf{X}_t) = f'(\mathbf{X}_t) \text{ for all } t < t_0\}. \quad (22)$$

In words, $\omega_B(\mathbf{X}_{t^*})$ asks how far apart two admissible counterfactual trajectories can be after treatment even though the pre-treatment data cannot distinguish between them. The definition describes uncertainty within the function class, independently of any particular estimator.

We next ask how it relates to the GP prediction (see [Kanagawa et al. \(2025\)](#) for a recent treatment of this GP–RKHS connection). As shown in [Section 3.2](#), the GP posterior mean is equivalent to a kernel ridge regression estimator in $\mathcal{H}_k$ ([Kimeldorf and Wahba, 1970](#)). It can also be written as a weighted combination of pre-treatment outcomes. This representation allows us to separate learning error into two parts. The first is the error that remains when the weighted pre-treatment function values fail to reconstruct the target at t^* . The second arises because the weights are applied to noisy observations rather than the underlying function values. The following proposition shows how the GP posterior variance accounts for both components. For simplicity, we state the result for the unit ball $B = 1$; the deterministic bound scales linearly with B .

Proposition 2 (GP posterior variance controls the learning error). *Suppose g lies in the unit ball $\mathcal{F}_1$, so that $g^* = g$, and suppose outcomes are observed with noise, $Y_t = g^*(\mathbf{X}_t) + \varepsilon_t$, where ε_t are mean-zero with variance $\sigma^2 > 0$. The GP posterior mean is the weighting estimator $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_t$ with weights $\mathbf{w} = (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{k}_{t^*}$, and its posterior variance is*

$$\mathbb{V}_{t^*} = k(\mathbf{X}_{t^*}, \mathbf{X}_{t^*}) - \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{k}_{t^*}.$$

Let $k_{t^*}^\perp = k(\cdot, \mathbf{X}_{t^*}) - \sum_t w_t k(\cdot, \mathbf{X}_t)$ be the residual of the test feature after its ridge reconstruction from the training-feature span $\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_t)\}$, coinciding with the orthogonal-projection residual onto $\mathcal{S}_{\text{pre}}$ as $\sigma^2 \rightarrow 0$. Then the learning error decomposes as

$$g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) = \underbrace{\left(g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t)\right)}_{\text{deterministic error}} - \underbrace{\sum_t w_t \varepsilon_t}_{\text{noise propagation}},$$

and its two sources are controlled by the two summands of the GP posterior variance:

$$\left|g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t)\right| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k}, \quad \text{Var}\left[\sum_t w_t \varepsilon_t\right] = \sigma^2 \|\mathbf{w}\|^2, \quad \underbrace{\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 + \sigma^2 \|\mathbf{w}\|^2}_{\text{GP posterior variance}} = \mathbb{V}_{t^*}.$$

Therefore, the first summand of $\mathbb{V}_{t^*}$ gives the squared worst-case bound on the deterministic error and the second gives the variance of the propagated noise.

A proof is given in Supplementary Material B. The residual norm $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$ measures how well the post-treatment input can be reconstructed from the weighted pre-treatment inputs in the kernel's function space. When this reconstruction is accurate, the pre-treatment observations place a strong restriction on the counterfactual prediction. When it is poor, functions that are difficult to distinguish from the observed history may imply substantially different outcomes after treatment. As the target moves farther beyond the observed support, this unrecovered component generally becomes larger, which makes the bound extrapolation-aware. The kernel determines which kinds of trajectories are easy for the model to represent. A smooth trend, for example, may have a small RKHS norm under a kernel designed to represent smooth trends, but a much larger norm under a poorly aligned kernel. Thus, the same trajectory can be simple under one kernel and complicated under another. The combined kernel of Section 3.2 is designed with this in mind, giving trend, seasonality, and local variation separate components in which they can be represented more naturally. Supplementary Material B develops this relationship formally and gives the corresponding source-condition refinement of this idea.

The result is particularly clear in the noiseless training case ($\sigma^2 = 0$), where the pre-treatment function values are observed without error. The remaining learning error comes entirely from what these observations fail to determine about the post-treatment point. For the unit RKHS ball,

Corollary B.1 (Supplementary Material B) gives

$$|g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k} = \sqrt{\mathbb{V}_{t^*}}, \quad (23)$$

where $\mathbb{V}_{t^*}$ denotes the GP posterior variance with $\sigma^2 = 0$. Thus, the GP posterior standard deviation bounds the gap between the best function in the assumed class and the function recovered from the pre-treatment data.

The bound is also sharp. To see its significance, consider any estimator that uses the same pre-treatment observations and evaluate its largest pointwise prediction error over the admissible function class. The minimax risk is the smallest worst-case error attainable among all such estimators. Supplementary Material B shows that the GP posterior mean achieves this risk. More generally, for $\mathcal{F}_B$,

$$\mathfrak{R}_B(\mathbf{X}_{t^*}) = B\|k_{t^*}^\perp\|_{\mathcal{H}_k} = \frac{\omega_B(\mathbf{X}_{t^*})}{2}, \quad (24)$$

where the last equality connects the minimax result to our definition of extrapolation uncertainty. Two admissible trajectories can agree at every pre-treatment input yet differ after treatment by as much as $\omega_B(\mathbf{X}_{t^*})$. Half of this largest disagreement is exactly the smallest worst-case error any estimator can guarantee. The GP posterior mean attains that limit, and its posterior variance characterizes the corresponding uncertainty. With noisy training outcomes, the uncertainty also reflects errors introduced by learning from noisy observations. Proposition 2 shows that the GP posterior variance contains both the deterministic reconstruction error and the variance of propagated noise. The proofs and further results are given in Supplementary Material B.

These results explain why GP regression is well suited to the extrapolation problem in ITS. The pre-treatment data may fit several plausible trajectories without telling us whether the untreated outcome would continue linearly or more wildly. The GP cannot determine which continuation is correct, but its kernel makes the assumed class of trajectories explicit, and its posterior variance provides a principled measure of uncertainty within that specification. Allowing more complex continuations, such as quadratic rather than only linear trends, may lead to greater uncertainty and weaker conclusions. The argument can also be read in reverse: if an estimated effect remains distinguishable from zero only under a kernel that restricts the counterfactual to a linear trend, the researcher must defend excluding more flexible alternatives. Comparing results across plausible

kernels therefore reveals which extrapolation assumptions are needed to sustain a causal conclusion.

5 Calibrated Simulation Studies

We now examine the finite-sample performance of the proposed estimator. Our main interest is whether its uncertainty intervals remain well calibrated as the counterfactual is extrapolated beyond the pre-treatment data.

Setup. We calibrate the DGPs to the monthly series in our empirical application (Section 6), which exhibits seasonality and an underlying trend. Point estimation is evaluated through bias and RMSE, with primary emphasis on interval coverage as the pre-treatment history grows. Untreated monthly outcomes are generated as $y_t(0) = g(t) + \varepsilon_t$ for $t = 1, \dots, n_{\text{pre}} + 12$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$. Treatment begins at $t_0 = n_{\text{pre}} + 1$, and a constant effect τ is added over the twelve-month post-treatment period. The two trend-based scenarios also include a small random Gaussian bump that introduces local deviations from their underlying trend functions.

Three scenarios are considered for g that share the monthly seasonality and short record of the application but become increasingly harder for the GP’s assumed function class to represent (see Supplementary Material C for more details). The first, *Smooth*, constructs the trajectory from a finite expansion of the combined kernel, providing a benchmark in which the DGP is closely aligned with the estimator. The GP still selects its own hyperparameters, however, and is not given the generating specification. The second, *Nonlinear trend*, combines relatively mild seasonality with a trend that initially grows but gradually approaches a plateau. The third, *Saturating trend*, combines a trend that flattens at both ends with a stronger annual cycle, requiring the estimator to extrapolate both the underlying trend and pronounced seasonal variation.

We fit the GP as in the application of Section 6 and compare it with three popular methods as benchmarks: (i) a segmented regression carrying a linear pre-trend and an annual harmonic, (ii) C-ARIMA (Menchetti et al., 2023), and (iii) CausalImpact, a Bayesian structural time-series model (Brodersen et al., 2015). For each scenario we let n_{pre} run from 12 to 120 months in steps of twelve, and draw 500 replications at each length. We report absolute bias and root-mean-squared

error (RMSE) to summarize point estimation, and 95% coverage to assess whether each method's intervals are calibrated over the post-treatment window.

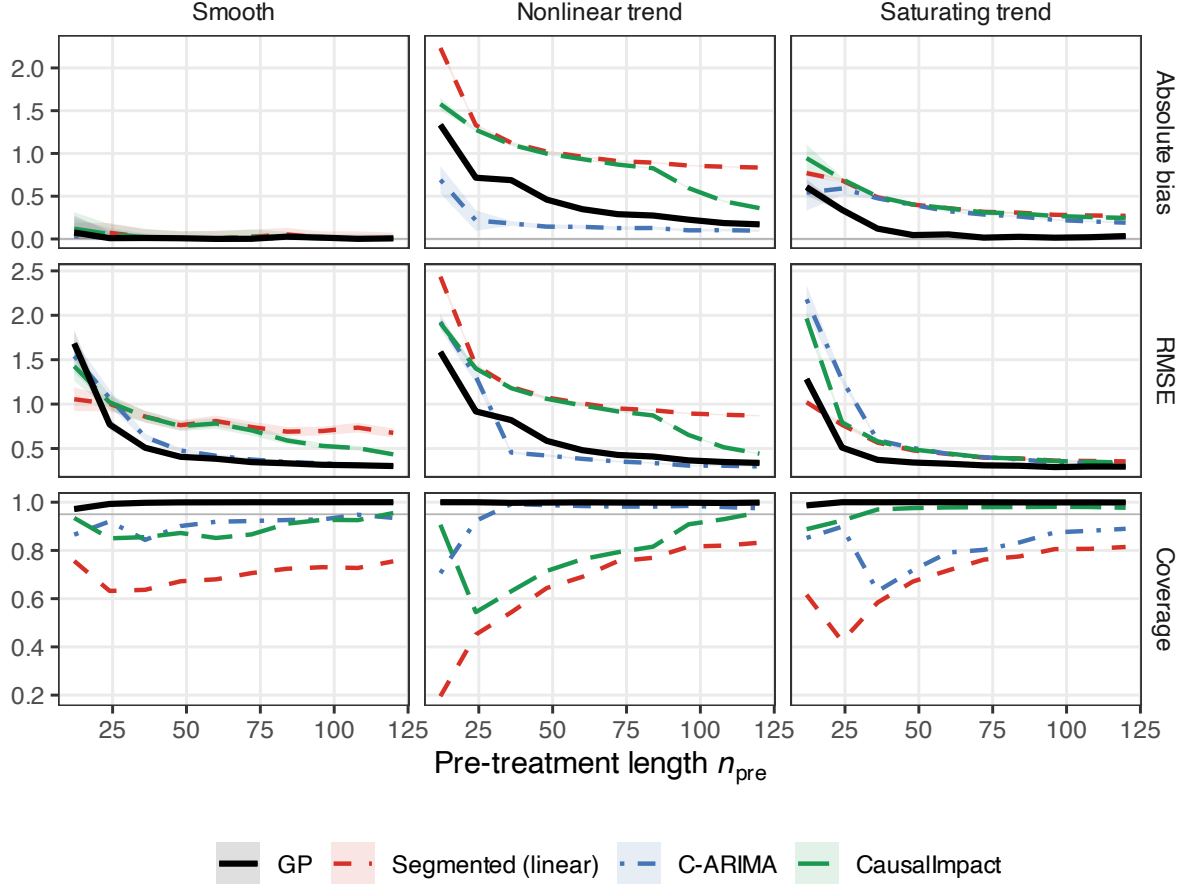


Figure 2. Absolute bias, RMSE and 95% coverage against the pre-treatment length n_{pre} , over a fixed twelve-month horizon and 500 replications, for three scenarios (columns). Bias is that of the average post-treatment effect; RMSE and coverage are computed at each post-treatment period and then averaged. The GP (black) is compared with a segmented regression, C-ARIMA and CausalImpact. Shaded bands are Monte-Carlo 95% intervals.

Results. Figure 2 reports absolute bias, RMSE, and coverage across the three scenarios and different pre-treatment lengths. The clearest differences between methods appear in coverage: the GP achieves at least the nominal 95% coverage in all three scenarios, even with only twelve pre-treatment observations. This finding is consistent with its ability to tame the extrapolation uncertainty, as elaborated in Section 4. The benchmark methods exhibit different coverage pat-

terns. Segmented regression falls substantially below nominal coverage, particularly when $Y(0)$ evolves nonlinearly. Its intervals account for uncertainty conditional on the linear model, but not for alternative continuations. **C-ARIMA**, with the seasonal period specified, approaches nominal coverage as the pre-treatment history grows, although it under-covers with shorter pre-treatment histories. **CausalImpact** shows more variable coverage across sample sizes, falling below nominal at intermediate lengths before recovering with additional pre-treatment observations. Point estimation improves as the pre-treatment record grows for every method. The methods differ much less in bias and RMSE than they do in coverage: the main advantage of the GP in these simulations is therefore not uniformly better point prediction, but more reliable uncertainty quantification across the extrapolation settings considered.

6 Empirical Application

Did the Supreme Court’s decision in *District of Columbia v. Heller* (2008) affect legal handgun purchases? On June 26, 2008, the Court recognized an individual right to keep and bear arms under the Second Amendment and struck down D.C.’s handgun restrictions. Although the ruling’s immediate regulatory effect was concentrated in D.C., its broader constitutional significance extended beyond the District. The decision established a major precedent concerning the individual right to bear arms and could have influenced expectations about future gun regulation and purchasing behavior nationwide, even though the application of the Second Amendment to the states was not established until *McDonald v. Chicago* (2010).

This makes *Heller* a useful setting for the design developed above. Other jurisdictions experienced no comparable immediate regulatory change, but their purchasing behavior may nevertheless have responded to the ruling. They therefore cannot simply be regarded as clean unaffected controls. The empirical question is therefore not simply treated versus control, but where and when the ruling produced a detectable change. If the unit-specific counterfactuals are well calibrated, we should see a large immediate effect in D.C. and little corresponding movement elsewhere.

We use monthly handgun background check rates per 100,000 population from the FBI’s National Instant Criminal Background Check System (NICS) as a proxy for handgun purchases ([Stein](#)

dley and Kosla, 2018). The panel includes all U.S. states and D.C. except Hawaii, whose record is incomplete, yielding $N = 50$ jurisdictions. The raw monthly series appears in Supplementary Material E. We restrict the post-treatment analysis to July–October 2008, ending before the November presidential election. This choice is motivated by the possibility that election-related concerns about future gun restrictions could influence purchases, making it more difficult to isolate the effect of *Heller*. Anticipation of the ruling itself presents a separate concern (Bertoli et al., 2026). In D.C., the near-total handgun ban limited opportunities for legal purchases in advance of the decision, lending some support to the no-anticipation assumption. Supplementary Material D.3 examines these identifying assumptions in greater detail.

Model. The monthly background check series exhibit seasonality as well as longer-run movements, motivating the Gaussian+Periodic+Linear kernel described in Section 3.2. We fit a separate GP for each jurisdiction, allowing the Gaussian component to capture local variation, the periodic component to represent annual seasonality ($p = 12$), and the linear component to accommodate longer-run trends. Month indicators are also included as covariates to account for month-specific baseline levels.

Illustrative Cases: D.C. and Washington State. We begin with D.C., where the ruling directly invalidated the handgun ban, and Washington state, whose pre-treatment trend and seasonal patterns closely resemble the national average. Figure 3 compares their fitted counterfactuals, post-treatment effects, and placebo estimates. The contrast is visible in both the fitted counterfactuals and the resulting effect estimates. In D.C., observed handgun checks rise sharply above the predicted counterfactual immediately after the ruling, producing positive estimated effects throughout the four-month post-treatment window. The placebo estimates remain near zero. Washington state shows no comparable departure: its post-treatment observations remain within the predicted counterfactual range, and the estimated effects are statistically indistinguishable from zero.

Localized *Heller* Effect and National Average. Figure 4 extends the analysis to all jurisdictions. The left panel presents their four-month cumulative effects, standardized by each

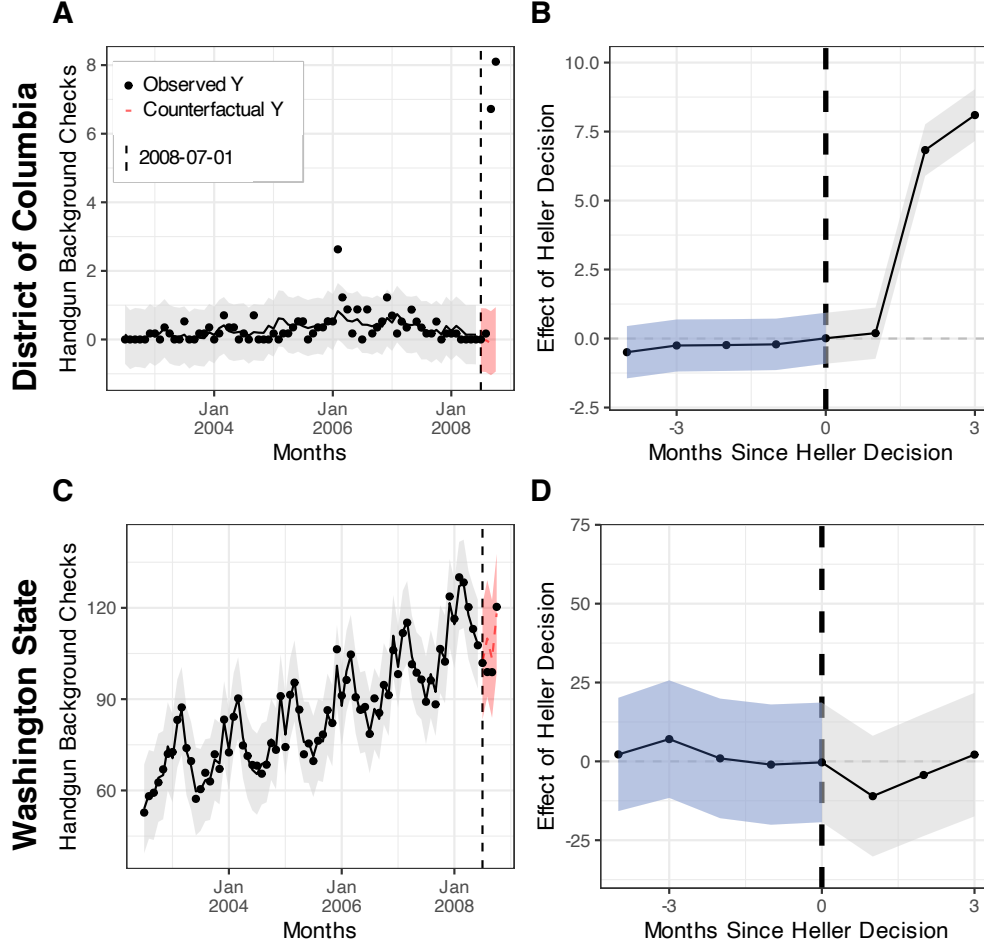


Figure 3. Counterfactual Predictions and Treatment Effects for D.C. and Washington State. *Note:* The top row shows D.C. and the bottom row Washington state. The left panels display observed outcomes (black dots), the fitted GP (gray bands), and post-treatment counterfactual predictions (red bands). The right panels show monthly treatment effects (gray) and placebo estimates (blue). Vertical dashed lines indicate the *Heller* decision.

jurisdiction’s pre-treatment variability to facilitate comparison across different baseline purchase levels. D.C. stands out with a large positive standardized effect, whereas the other jurisdictions cluster around zero. The standardized magnitude for D.C. requires caution, however. Its handgun ban kept pre-treatment checks close to zero, leaving little variation against which to scale the estimated effect. We therefore report its magnitude in the original units: the cumulative four-month effect is 15.1 additional checks per 100,000 population, with a 95% interval of [13.0, 17.3], equivalent to roughly 90 additional background checks.

The right panel traces the national average effect $\hat{\tau}_{t*}$ over time. It remains close to zero across the

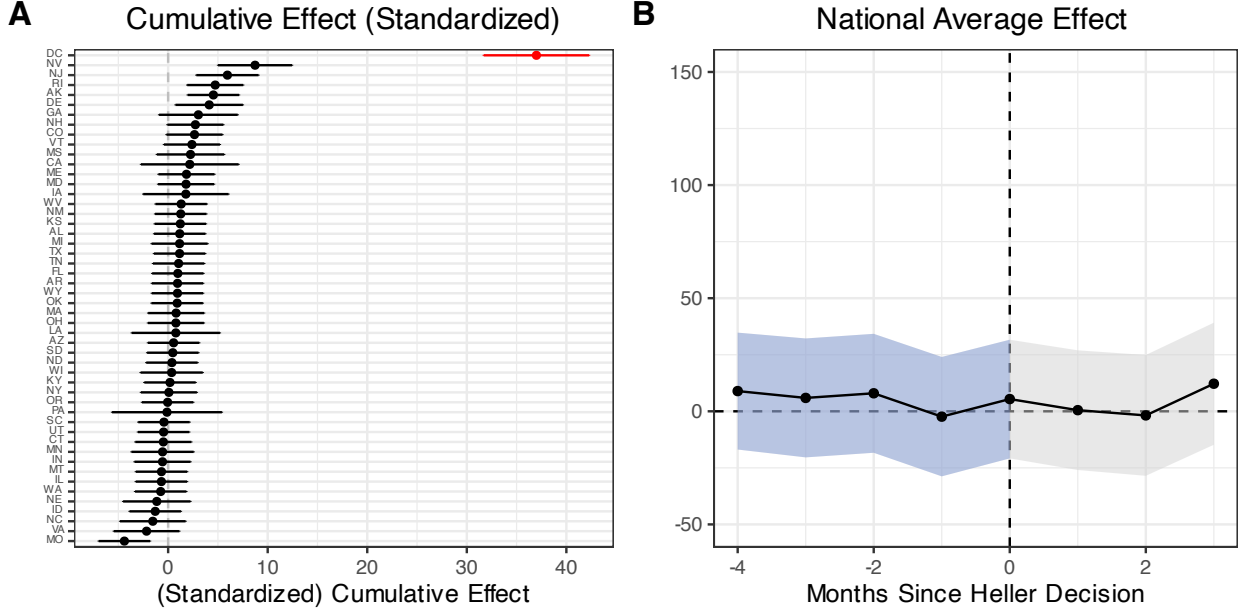


Figure 4. Heterogeneous and Nationwide Effects of the *Heller* Decision on Handgun Purchases. *Note:* The left panel gives unit-specific standardized cumulative effects for the four months following the decision, with 95% prediction intervals. The right panel gives the estimated national average treatment effect $\hat{\tau}_{t*}$ by month.

post-treatment months, and the cumulative version in Supplementary Material E shows the same pattern. Together with the jurisdiction-specific estimates, this suggests that the observed increase was concentrated in D.C. rather than broadly shared across the country. The national results also provide context for the concern about election-related purchasing. A widespread response to the approaching election could have produced increases outside D.C., yet we find little corresponding movement in the national average during July–October. This pattern is less consistent with a broad nationwide increase, although it cannot rule out election-related behavior specific to D.C. or other concurrent influences on its purchases.

We further assess the counterfactual predictions using the one-step-ahead placebo check of Section 3.3. Segmented regression excludes zero in 58–67% of pre-treatment periods, indicating apparent effects where treatment had not occurred, whereas the GP excludes zero in none. The contrast mirrors the simulations of Section 5, namely, segmented-regression intervals remain conditional on the chosen trend specification and do not account for alternative trajectories that fit the pre-treatment record similarly well but diverge out of sample. The GP placebo results are therefore reassuring about its pre-treatment forecasting performance, although they cannot prove that iden-

tifying assumptions hold. Additional diagnostics support the stability of the reported estimates. In-sample residuals show no visible deterioration as the treatment date approaches, and sensitivity analysis across a broad range of length-scale values b yields similar estimates for the D.C. effect and the national average (Supplementary Materials E and F).

We also examine long gun checks as a placebo outcome, since the ruling directly invalidated D.C.’s handgun ban rather than a comparable restriction on long guns. The standardized cumulative effect in D.C. is 0.25, with a 95% interval of $[-5.2, 5.7]$, providing no evidence of a comparable change in this outcome. Supplementary Material E presents the cumulative effects, placebo comparisons, residual diagnostics, and long gun results in greater detail.

7 Relation to Existing Approaches

Counterfactual outcomes can be learned from contemporaneous comparisons across units, from a unit’s own history over time, or from some combination of the two. When treatment occurs simultaneously across all units, the first source of information is no longer available, making temporal extrapolation central to the problem. Methods applicable in this setting therefore differ mainly in how they project the pre-treatment history forward and in how they quantify the uncertainty that comes with doing so. We first compare the GP to time-series methods designed for this task and then connect its identifying logic to related causal inference methods.

Time Series Approaches. Classical ITS methods, from the intervention analysis of [Box and Tiao \(1975\)](#) to segmented regression, typically specify both a model for the untreated trajectory and a treatment response pattern, such as a step, pulse, or gradual change. These choices facilitate estimation and interpretation but place restrictions on the trajectories considered. More recent approaches relax aspects of this structure: within a potential outcomes framework, [Mira-trix \(2022\)](#) simulate counterfactual trajectories from a linear prediction model, while [Felton \(2023\)](#) imputes counterfactuals from ARIMA fits. Our approach likewise learns the counterfactual from pre-treatment history, but specifies its functional structure through a covariance kernel rather than a linear or autoregressive model. The GP also provides a measure of predictive uncertainty whose

relationship to extrapolation error is characterized in Section 4.

The **CausalImpact** method (Brodersen et al., 2015) uses a Bayesian structural time-series model with latent state components. As Hazlett and Xu (2018) emphasize, its counterfactual can be expressed as a linear function of prior outcomes. The GP shares this feature: its posterior mean is also a linear smoother, $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_{it}$. The distinction therefore lies not in linearity with respect to observed outcomes, but in the assumptions governing the underlying trajectory. The GP kernel determines the prediction weights and the covariance structure linking pre- and post-treatment inputs. Its posterior variance also responds directly to how far the post-treatment target lies from the information contained in the pre-treatment inputs, which is the extrapolation property developed in Section 4. **C-ARIMA** (Menchetti et al., 2023) similarly learns the untreated process from pre-treatment data and projects it forward, rather than estimating a prespecified intervention effect through a regression with ARIMA errors fitted to the full series. It models the outcome’s dynamics through autoregressive, differencing, and moving-average components. The GP follows the same counterfactual forecasting strategy but specifies the dependence structure through a covariance kernel rather than ARIMA dynamics. Both approaches quantify forecast uncertainty under their respective assumptions; our RKHS analysis additionally connects the GP posterior variance to worst-case learning error within the chosen function class.

Related Identification Strategies. DiD and its modern variants rely on changes in untreated comparison units to estimate how treated outcomes would have evolved absent treatment, under the parallel trends assumption. Synthetic control instead constructs a weighted combination of untreated donor units that reproduces the treated unit’s pre-treatment trajectory, relying on that relationship to persist after treatment. Under simultaneous treatment, neither approach applies because contemporaneous untreated units are unavailable. Our approach retains the underlying idea of learning a counterfactual relationship before treatment, but does so within each unit rather than across units. Mean Sufficiency restricts the role of unobserved factors in the conditional mean, while the kernel determines how the relationship learned from pre-treatment history is extended to post-treatment inputs. Temporal extrapolation thus replaces the missing cross-unit comparisons.

RDiT exploits the local comparability of observations immediately before and after the treat-

ment cutoff to estimate an effect at the event time. This local focus, however, does not by itself reveal how effects emerge, persist, or fade over a longer post-treatment period. Our framework instead extends the pre-treatment trajectory to each post-treatment period, using the kernel to make assumptions about temporal comparability explicit and the GP posterior variance to quantify uncertainty as the counterfactual is extrapolated. RDiT also faces particular challenges with discretely observed outcomes (Bertoli et al., 2026; De la Cuesta and Imai, 2016), which the GP accommodates directly.

8 Concluding Remarks

Universal treatment removes the contemporaneous controls on which difference-in-differences and synthetic control rely, leaving an ITS design that must extrapolate the counterfactual from each unit’s own history. We propose GP regression as a flexible approach to this problem, with the kernel making assumptions about the untreated trajectory explicit and the posterior variance quantifying uncertainty under those assumptions. Our identification analysis distinguishes what the pre-treatment data reveal from what must be extrapolated, while the RKHS results connect the GP posterior variance to worst-case learning error and extrapolation uncertainty within the chosen function class. Simulations demonstrate reliable interval coverage across the scenarios considered, and the empirical applications illustrate how the method can trace dynamic effects without imposing a fixed treatment response pattern. An open-source R implementation accompanies the paper.

Several limitations deserve emphasis. Performance depends on the kernel and its implied function class, particularly beyond the observed support. The worst-case learning bound also depends on the RKHS norm of the target, $\|g^*\|_{\mathcal{H}_k}$, which cannot generally be estimated from data. Fitting each series on its own forgoes the efficiency that cross-sectional structure could supply, which hierarchical or multi-task formulations (Ben-Michael et al., 2023) could recover for short panels at the cost of assumptions about cross-unit similarity. Computation is a further constraint, since inverting the kernel matrix scales poorly beyond a few thousand observations; sparse approximations (e.g., Titsias, 2009) may extend the method to long daily series.

Two extensions merit exploration. The distance-based covariance accommodates irregularly spaced observations without modification, a common feature of administrative and judicial records; our applications use evenly spaced series, and a formal study of performance under missing observations would be valuable. More broadly, the logic developed here applies wherever a credible counterfactual must be extrapolated from a unit’s own history and the conclusions drawn from it must remain extrapolation-aware.

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